

VALVE SPRING DRAWING COMPARISON

Exhaust Spring A1770530600 vs Intake Spring A1770530500

Mercedes-Benz AMG M177 Engine · Scherdel GmbH · DIN 2098 Series

Wire Section	Free Length Δ	Seat Load Δ	Full-Lift Load	Avg. Work Rate
2.92 × 3.66 mm oval (shared)	Exhaust 47.0 mm Intake 46.1 mm (+0.9 mm)	Exhaust 270 N Intake 250 N (+8 %)	Both 620 N (shared)	Exhaust 35.0 N/mm Intake 37.0 N/mm

Engineering Drawings

Exhaust Valve Spring A1770530600

Technical specifications table (left):

Item	Description	Value	Unit
1	Wire diameter	2.92	mm
2	Inner diameter	3.66	mm
3	Outer diameter	6.62	mm
4	Inner diameter	3.66	mm
5	Outer diameter	6.62	mm
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99	Outer diameter	6.62	mm
100	Inner diameter	3.66	mm

Material properties table (right):

Property	Value	Unit
Yield strength	1020	N/mm²
Tensile strength	1180	N/mm²
Elongation	10	%
Reduction of area	40	%
Impact energy	270	J
Hardness	217	HV

Intake Valve Spring A1770530500

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A 177 053 06 00

Valve Spring – Exhaust (Auslassfeder)

A 177 053 05 00

Valve Spring – Intake (Einlassfeder)

Parameter Comparison Table

N o.	Symbol / Parameter	Unit	Exhaust A1770530600	Intake A1770530500	Δ / Remark
1	d — wire diameter (oval cross-section)	mm	2.92 × 3.66	2.92 × 3.66	Identical
2	Dio — inner dia. top (oben)	mm	12.00 ± 0.20	12.00 ± 0.20	Identical
3	Diu — inner dia. bottom (unten)	mm	15.90 ± 0.20	15.90 ± 0.20	Identical
4	Deo — outer dia. top	mm	19.32	19.32	Identical
5	Deu — outer dia. bottom	mm	23.22	23.22	Identical
6	# — spring index	—	5.32	5.32	Identical
7	Coiling direction	—	Right (RH)	Right (RH)	Identical
8	G — shear modulus	N/mm ²	79 500	79 500	Identical
9	L0 — free length	mm	47.0	46.1	Exhaust +0.9 mm
10	L1 — installed length	mm	36.10	36.10	Identical
11	L2 — max. working length	mm	26.10	26.10	Identical
12	Lc — solid length	mm	24.60 – 0.9	24.60 – 0.9	Identical
13	F1 — load at L1 (seat/installed)	N	270 ± 13	250 ± 12	Exhaust +20 N (+8 %)
14	F2 — load at L2 (full lift)	N	620 ± 27	620 ± 21	Same load; tighter tol. on intake
15	k_avg — avg. working rate (derived)	N/mm	35.0	37.0	Exhaust softer over stroke
16	τ_k1 — torsional stress at L1	N/mm ²	453	419	Exhaust +34 N/mm ² (+8 %)
17	τ_k2 — torsional stress at L2	N/mm ²	1 040	1 040	Identical
18	τ_kh — fatigue / stress range	N/mm ²	587	521	Exhaust +66 N/mm ² (+13 %)
19	nt — total coils	—	8.6	8.6	Identical
20	na — active coils	—	4.4 – 3.4	4.4 – 3.4	Identical
21	e_min — min. coil spacing	mm	0.55	0.55	Identical
22	Spring shape	—	Beehive	Beehive	Identical
23	End treatment	—	Closed & ground 240°	Closed & ground 240°	Identical
24	Shot peening	—	Yes	Yes	Identical
25	Colour stripe 1	—	Violet RAL 4002	Violet RAL 4002	Identical — spring type marker
26	Colour stripe 2 (ID colour)	—	Brown RAL 8004	Yellow RAL 1006	DIFFERENT — position identifier
27	Surface protection	—	Wax	Wax	Identical
28	Heat treatment	—	Yes (DIN 7386.20)	Yes (DIN 7386.20)	Identical
29	Supplier / std.	—	Scherdel / DIN 2098	Scherdel / DIN 2098	Identical
30	Material	—	VD SiCrNiV SC	VD SiCrNiV SC	Identical (DIN 17 223)

★ Highlighted rows indicate parameters that differ between the two springs.

Engineering Analysis

1. Shared Design Basis

Both springs are manufactured by Scherdel GmbH to DIN 2098 / DIN 17 223 standards using **VD SiCrNiV SC** valve-spring steel wire with an **oval cross-section (2.92 × 3.66 mm)**. The beehive (Bienenkorb) profile, coiling direction (RH), spring index (5.32), coil count ($n_t = 8.6$), end treatment (closed-and-ground, 240°), solid length ($L_c = 24.60$ mm), installed length ($L_1 = 36.10$ mm), maximum working length ($L_2 = 26.10$ mm), and full-lift force ($F_2 = 620$ N) are all identical. Both springs receive the same shot-peening, wax surface protection, and heat treatment. This shared basis minimises part proliferation and tooling cost — the two parts diverge only where the valve dynamics demand it.

2. Free Length and Preload Difference

The sole geometric difference between the two springs is free length: **$L_0 = 47.0$ mm (exhaust) vs 46.1 mm (intake)**, a delta of 0.9 mm. Because both springs are installed to the same assembly height ($L_1 = 36.10$ mm), the 0.9 mm extra free length compresses the exhaust spring 0.9 mm further at installation, raising the **seat load (F1) from 250 N to 270 N — an 8 % increase**. The torsional stress at the installed length rises proportionally: 453 N/mm² (exhaust) vs 419 N/mm² (intake).

3. Derived Spring Rates Over the Working Stroke

Using the simple average-rate formula $k = \Delta F / \Delta L$ over the 10 mm working stroke ($L_1 - L_2$):

Exhaust: $k = (620 - 270) / (36.10 - 26.10) = 35.0$ N/mm

Intake: $k = (620 - 250) / (36.10 - 26.10) = 37.0$ N/mm

Both springs reach the same force (620 N) at full valve lift, but the exhaust spring starts from a higher seat load, so its effective spring rate over the working stroke is slightly **lower (35.0 vs 37.0 N/mm)**. The intake spring therefore has approximately **5.7 % higher dynamic stiffness** across the lift event. Note that the beehive geometry produces a naturally progressive rate; these figures represent mean values across the stroke.

4. Fatigue Stress Range and Durability

The fatigue stress range $\tau_{kh} = \tau_{k2} - \tau_{k1}$ is the primary HCF driver for valve springs. The exhaust spring carries a **13 % higher fatigue range (587 vs 521 N/mm²)**. This is consistent with the higher preload and reflects the harsher thermal and pressure environment of the exhaust side:

- **Exhaust valves** are exposed to residual combustion gases during the overlap period, and exhaust-side combustion pressure acts against the valve in early exhaust stroke. A higher seat load ensures positive valve closure despite back-pressure.
- **Intake valves** benefit from lower back-pressure — only intake-manifold vacuum assists closure — so a lower seat load is sufficient while keeping fatigue stress range within limits.

Both springs share the same peak stress at full lift ($\tau_{k2} = 1\,040$ N/mm²), confirming the valvetrain is dimensioned to the same maximum stress level; it is only the *starting point* of each load cycle that differs.

5. Load Tolerance and Assembly Quality

The full-lift tolerance on the exhaust spring is **±27 N** vs **±21 N** for the intake spring. A tighter tolerance on the intake spring may reflect closer attention to intake air-flow symmetry and cylinder-to-cylinder variation, whereas the exhaust spring relies more on the higher absolute preload level to guarantee closure. Both are well within the limits prescribed by DIN 2098 Part 2 (Gütevorschrift) and subject to statistical process control (Mitteldruckprüfen, DIN 6515 escalation level C, verified by Scherdel).

6. Colour Coding and Field Identification

Both springs share the first colour stripe (violet RAL 4002) indicating the spring *type* (beehive, oval wire, AMG series). The **second colour stripe uniquely identifies the position**:

- Exhaust: **brown (braun) RAL 8004**
- Intake: **yellow (gelb) RAL 1006**

This two-colour coding system prevents incorrect installation during engine assembly or service — the springs are externally identical in shape, and only the colour stripe distinguishes them without measurement.

Conclusions

- **Near-identical springs, one key difference:** The exhaust spring is the same part as the intake spring except for a 0.9 mm longer free length, which raises the seat load by 20 N (8 %) and the fatigue stress range by 66 N/mm² (13 %).
- **Same peak force at full lift (620 N):** Both springs are designed to deliver the same maximum force at full valve lift. This ensures the valvetrain cam/follower system, designed to a single peak force level, can be applied to both banks without modification.
- **Higher exhaust preload is physically justified:** The additional seat load on the exhaust spring (270 vs 250 N) compensates for combustion back-pressure on the exhaust valve, preventing valve float or delayed closure under high-RPM or high-load conditions.
- **Intake spring has slightly higher working stiffness:** Paradoxically, the softer (lower-preload) intake spring shows a 5.7 % higher average working rate (37.0 vs 35.0 N/mm) over the valve lift event, because it must traverse the same force range from a lower starting point.
- **Fatigue dimensioning is consistent:** Both springs share the same peak torsional stress at full lift (1 040 N/mm²), confirming unified fatigue dimensioning. The exhaust spring operates at a higher mean stress, which in a Goodman diagram reduces its fatigue allowance; the VD SiCrNiV SC material and shot-peening are chosen to accommodate this.
- **Colour coding is critical for correct assembly:** The brown vs yellow second stripe is the only visual distinction between the springs. Swapping them would install too-high preload on intake and too-low on exhaust — causing potential intake-side over-stress and exhaust-side valve float.
- **No changes to manufacturing process or supplier:** Both parts share supplier (Scherdel), material specification, end treatment, surface protection, and quality standard. The free-length difference is the sole production distinguisher, minimising supply-chain complexity.

Summary: The exhaust and intake valve springs of the M177 engine are nearly identical beehive springs sharing all geometry and manufacturing parameters. The single engineering difference — a 0.9 mm longer free length on the exhaust spring — deliberately raises the seat load to resist exhaust back-pressure, while both springs converge to the same 620 N force at maximum valve lift. Correct installation is assured through a distinctive colour-stripe identification system (brown = exhaust, yellow = intake).

Document prepared automatically from engineering drawings A1770530600-3 and A1770530500-4. Derived calculations use nominal drawing values. All specifications subject to Scherdel DIN 2098 Part 2 quality protocol.